



Cloud Management Concepts

INE Cloud Associate Series





Tracy Wallace

Azure Solutions Architect
Expert



twallace@ine.com



@TracyWallaceTec



linkedin.com/in/tracy-wallace-746482a



Topics

Management Foundations
Resource Management

Learning Objectives

After completing this course, you will be able to:

- Describe the unique aspects of managing resources in the cloud
- Identify the cloud resource management tasks
- Explain the fundamentals of cloud access control

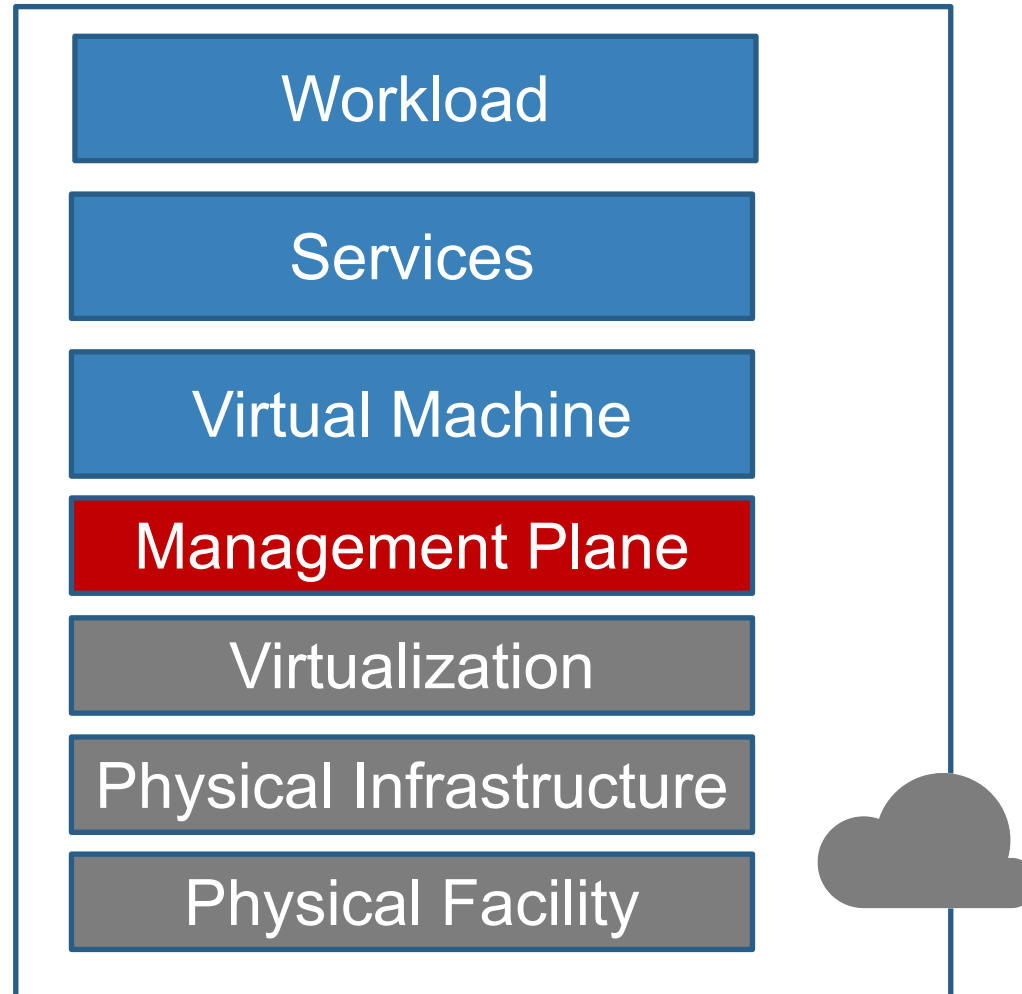


Cloud Shared Responsibility Model

Cloud Shared Responsibility Model

- + Shared Responsibility
- + Management Responsibility
- + Security Responsibility
- + Resiliency Responsibility
- + Workload Responsibility
- + Demo: Management

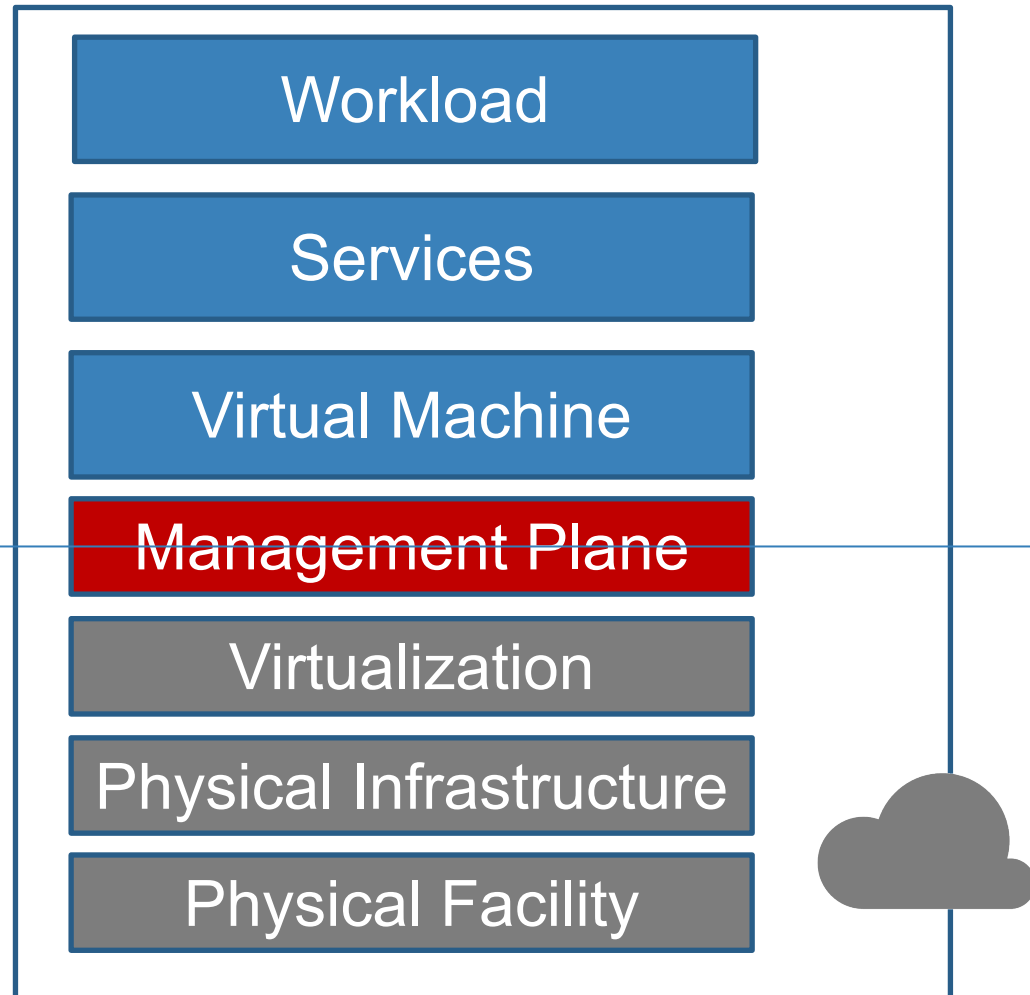
Shared Responsibility



Management Responsibility

You are always responsible for managing your Azure AD tenant, and access to your subscription

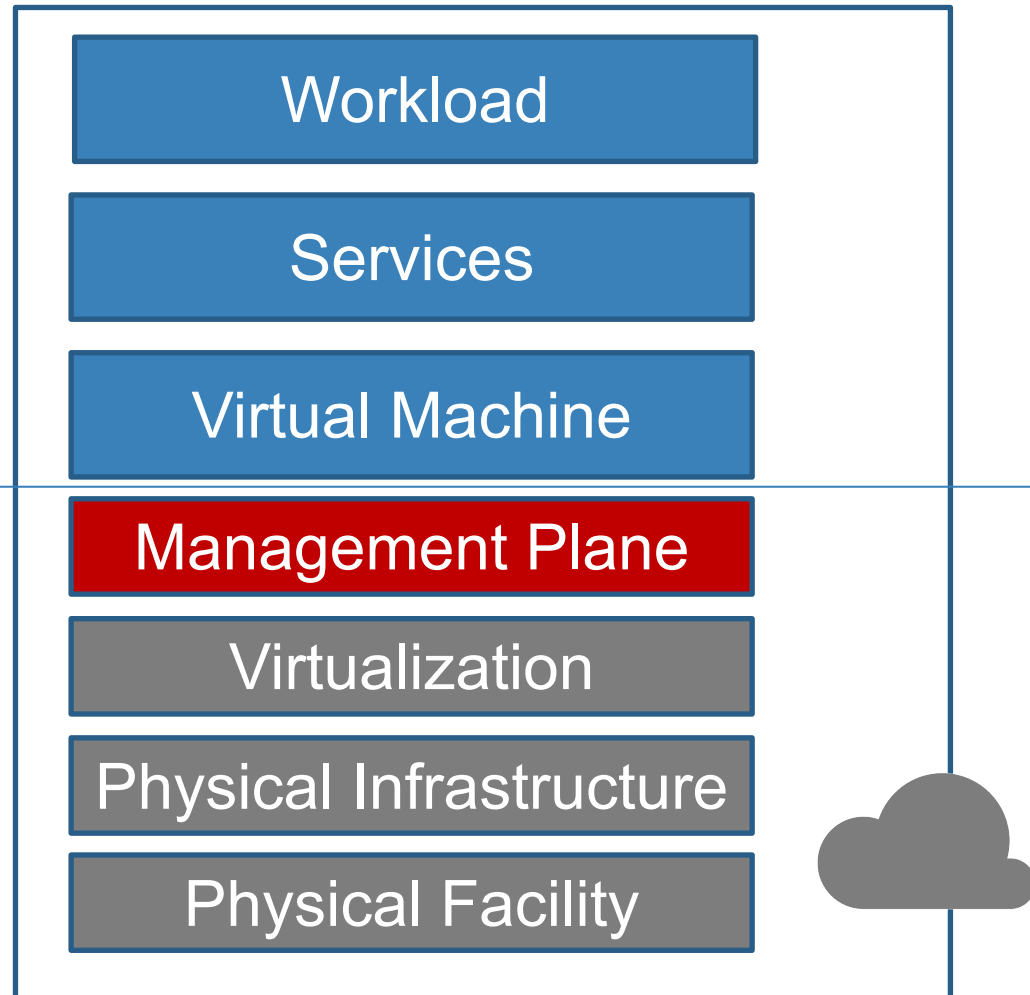
The cloud provider is always responsible for managing and securing the facility, infrastructure, virtualization, and cloud management plane



Management Responsibility

Infrastructure as a Service

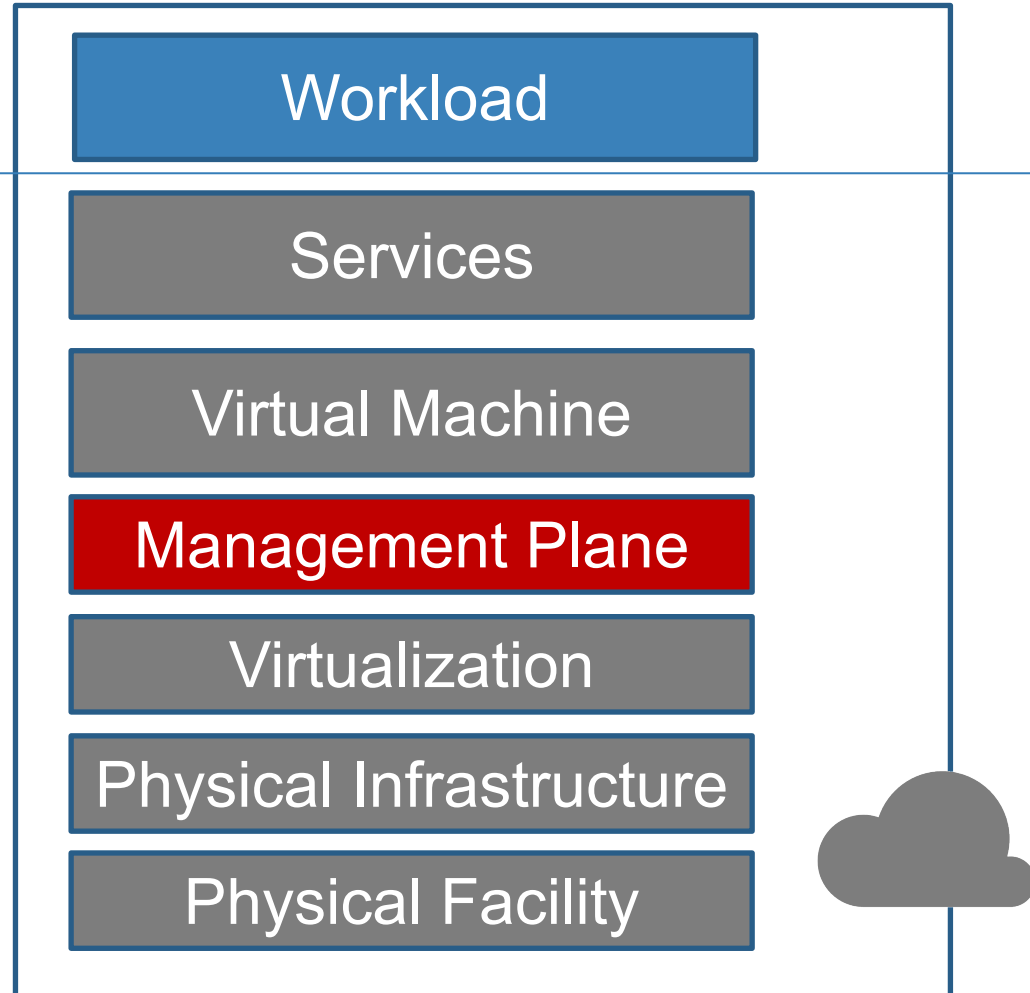
You are responsible for managing the virtual machine (operating system) and above



Management Responsibility

Platform as a Service

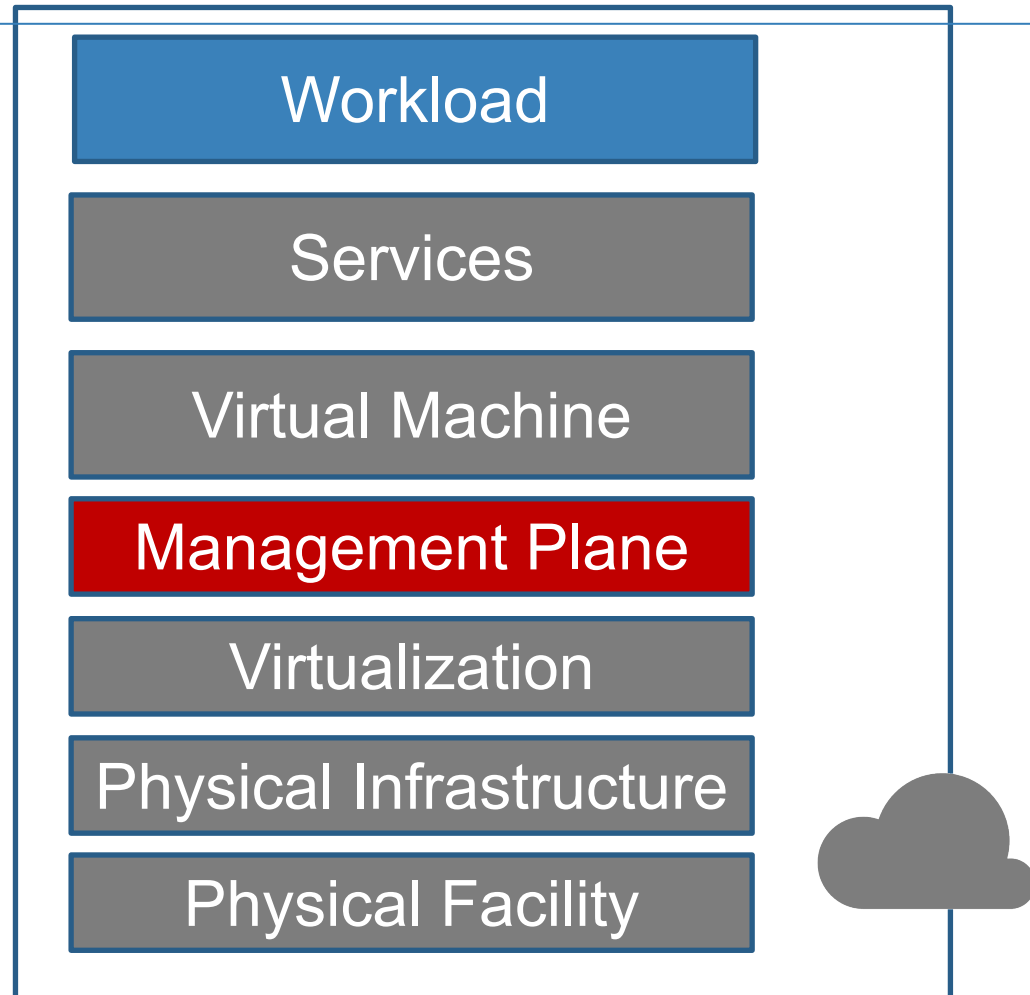
You are responsible for managing the workload



Management Responsibility

Software as a Service

The provider is responsible for managing the full stack. You may be responsible for managing customizations. You are still responsible for managing usage and adhering to good practices and compliance.



Security Responsibility

- Provider responsibility
 - + Physical security
 - + Infrastructure security
 - + Platform security
 - + Identity system security
 - + Standards compliance
- Customer responsibility
 - + Identity security
 - + Data security
 - + Application security
 - + Standards compliance

Resiliency Responsibility

- 99.99% Uptime service level agreement (SLA)
- Build redundancy to support availability
- Build resiliency into applications

Workload Responsibility

- Workloads still fail in the cloud
- SaaS out of the box
 - + Workload failures are the responsibility of provider
 - + No customization
- Good software lifecycle management practices are still crucial

Demo: Management

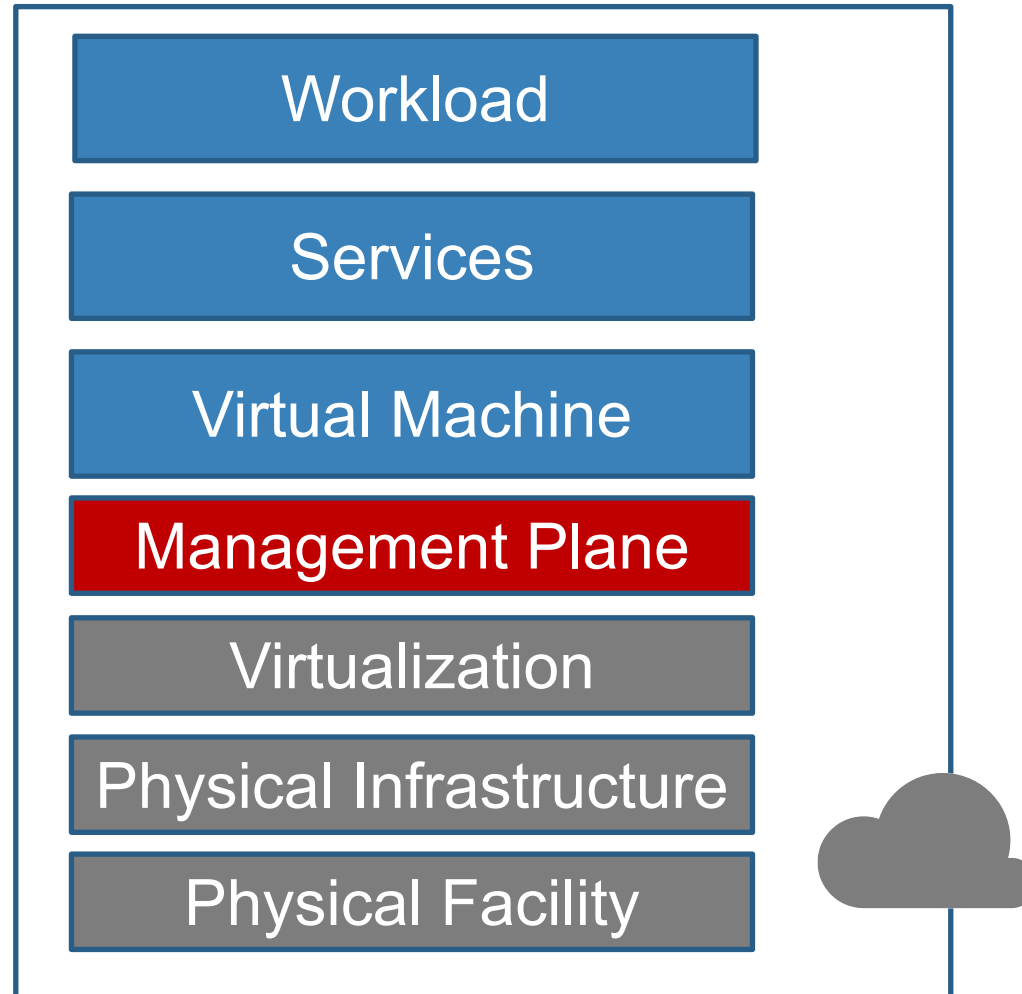


Cloud Management and Data Planes

Cloud Management and Data Planes

- + Cloud Control Planes
- + Control Plane
- + Data Plane
- + Demo: Data Plane Management

Cloud Control Planes



Management Plane

- Management plane is "the cloud"
 - + REST API
 - + Web-based console
 - + Command line tools

EC2 Dashboard New

Events

Tags

Limits

▼ Instances

Instances New

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Scheduled Instances

Resources

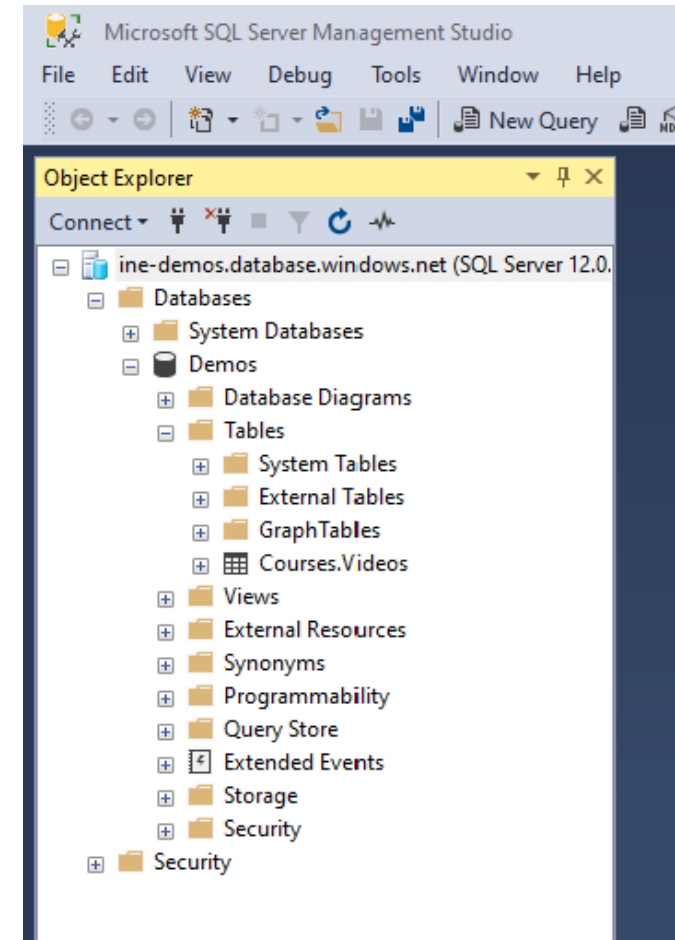
You are using the following Amazon EC2 resources in the US East (N. ')

Instances (running)	1	Dedic
Key pairs	3	Load
Snapshots	3	Volur

 Easily size, configure, and deploy Microsoft SQL Server Always

Data Plane

- Data plane is your workload
 - + Virtual machine OS and above for IaaS
 - + Data
 - + Applications
 - + Services



Demo: Data Plane Management



Managing Cloud Assets

Managing Cloud Resources

- + Provisioning Resources
- + Configuring Resources
- + Maintaining Resources
- + Demo: Resource Maintenance
- + Monitoring Resources
- + Change Management

Provisioning Resources

 Services ▾

Search for services, features, marketplace products, and docs [Alt+S]

twallace @ aws-wallace ▾ N. Virginia ▾ Support ▾

1. Choose AMI 2. Choose Instance Type 3. Configure Instance 4. Add Storage 5. Add Tags 6. Configure Security Group 7. Review

Step 1: Choose an Amazon Machine Image (AMI)

[Cancel and Exit](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace; or you can select one of your own AMIs.

Search for an AMI by entering a search term e.g. "Windows" ×

Search by Systems Manager parameter

Quick Start

My AMIs

AWS Marketplace

Community AMIs

☐ Free tier only ⓘ



Amazon Linux
Free tier eligible

Amazon Linux 2 AMI (HVM), SSD Volume Type - ami-0be2609ba883822ec (64-bit x86) / ami-0c582118883b46f4f (64-bit Arm)

Select

Amazon Linux 2 comes with five years support. It provides Linux kernel 4.14 tuned for optimal performance on Amazon EC2, systemd 219, GCC 7.3, Glibc 2.26, Binutils 2.29.1, and the latest software packages through extras. This AMI is the successor of the Amazon Linux AMI that is approaching end of life on December 31, 2020 and has been removed from this wizard.

Root device type: ebs Virtualization type: hvm ENA Enabled: Yes



Mac

macOS Catalina 10.15.7 - ami-01e2686a21e3bdf3f

Select

The macOS Catalina AMI is an EBS-backed, AWS-supported image. This AMI includes the AWS Command Line Interface, Command Line Tools for Xcode, Amazon SSM Agent, and Homebrew. The AWS Homebrew Tap includes the latest versions of multiple AWS packages included in the AMI.

Root device type: ebs Virtualization type: hvm ENA Enabled: Yes



Mac

macOS Mojave 10.14.6 - ami-0c71baab7b360e10a

Select

The macOS Mojave AMI is an EBS-backed, AWS-supported image. This AMI includes the AWS Command Line Interface, Command Line Tools for Xcode, Amazon SSM Agent, and Homebrew. The AWS Homebrew Tap includes the latest versions of multiple AWS packages included in the AMI.

Root device type: ebs Virtualization type: hvm ENA Enabled: Yes

1 to 41 of 41 AMIs

< >



Configuring Resources

inecosmos | Firewall and virtual networks

Azure Cosmos DB account

Search (Ctrl+/)

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Quick start
- Notifications
- Data Explorer

Settings

- Features
- Replicate data globally
- Default consistency
- Backup & Restore
- Firewall and virtual networks
- Private Endpoint Connections

Allow access from

☐ All networks ☒ Selected networks

Configure network security for your Azure Cosmos DB account. [Learn more.](#)

Virtual networks

Secure your Azure Cosmos DB account with virtual networks. [+ Add existing virtual network](#) [+ Add new virtual network](#)

Virtual Network	Subnet	Address range	Endpoint Status	Resource Group	Subscription
No network selected.					

Firewall

Add IP ranges to allow access from the internet or your on-premises networks. [+ Add my current IP \(75.183.242.53\)](#) ⓘ

IP (Single IPv4 or CIDR range)

No IP range filter configured.

Exceptions

- ☐ Accept connections from within public Azure datacenters ⓘ
- ☒ Allow access from Azure Portal ⓘ

Maintaining Resources

- Maintaining code base
- Maintaining data
- Maintaining security

Demo: Resource Maintenance

Monitoring Resources

 Services ▾

[Alt+S]

twallace @ aws-wallace ▾ N. Virginia ▾ Support ▾

CloudWatch

Dashboards

Alarms

0 ALARM

0 INSUFFICIENT

0 OK

Billing

Logs

Log groups

Insights

Metrics

Explorer **NEW**

Events

Rules

Event Buses

ServiceLens

Service Map

Traces

Container Insights **NEW**

Resources

 Update

Create cross-account cross-region dashboards for a complete view of your application. [Learn more](#)

CloudWatch: Overview ▾

1h 3h 12h 1d 3d 1w custom ▾

Actions ▾

 ▾

All resources ▾

Alarms by AWS service 

Services

Status	Alarm	Insufficient	OK
? EC2	-	-	-
? Elastic Block Store	-	-	-
? S3	-	-	-
? Usage	-	-	-

Recent alarms  [View recent alarms dashboard](#)Recent alarms will appear here.
[Learn more about CloudWatch Alarms.](#)

Change Management

- It is easy to change things in the cloud
- Governance is critical in the cloud
- Tools are available for managing change – platform specific and vendor neutral



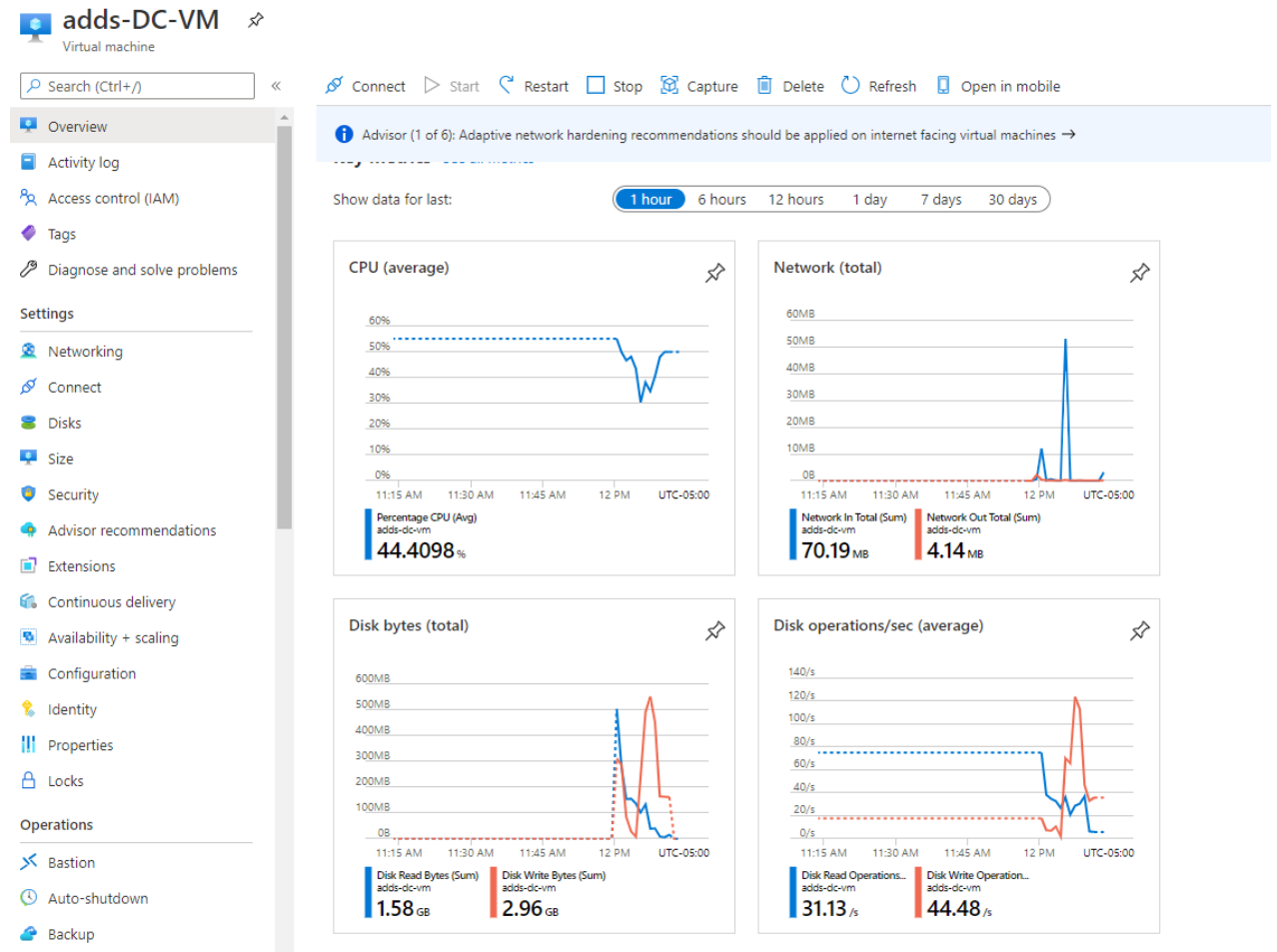
Monitoring and Alerts

Monitoring and Alerts

- + Cloud Monitoring
- + Unified Monitoring
- + Proactive Resource Management
- + Demo: Cloud Monitoring

Cloud Monitoring

- Resource Monitoring
- System Monitoring
 - + Azure Monitor
 - + AWS CloudWatch
 - + Google Cloud Monitoring



Unified Monitoring

- Nagios
- Splunk
- PRTG

Proactive Resource Management

- Cloud Automation
- Cloud Alerting

Choose launch template or configuration [Info](#)

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group. If you currently use launch configurations, you might consider migrating to launch templates.

Name

Auto Scaling group name
Enter a name to identify the group.

Must be unique to this account in the current Region and no more than 255 characters.

Launch template [Info](#)

Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

▼

[Create a launch template](#) [↗](#)

Version
 ▼

[Create a launch template version](#) [↗](#)

Description	Launch template	Instance type
demo launch template for auto-scaling	test-launch ↗ lt-06729bc5a4f0a3956	t2.micro
AMI ID	Security groups	Request Spot Instances
ami-0915bcb5fa77e4892	-	No

Demo: Cloud Monitoring



Cloud Identity and Access Management

Cloud Identity and Access Management

- + Cloud-Based Identity
- + Access Management
- + Demo: Cloud Access Management

Cloud-Based Identity

Access Management

- Users, Groups, Roles
- Federated users

Azure

AWS

GCP

Azure
AWS
GCP

Access Management

- Users, Groups, Permissions
- Federated users
- Policies grant permissions
 - Effect (allow/deny)
 - Action
 - Resources
 - Conditions

Azure
AWS
GCP

Access Management

- Users, Groups, Roles, Policies
- Federated users (preview as of Feb-2021)
- Policies link users to roles on resources
 - User
 - Role
 - Resource
 - Condition

Demo: Cloud Access Management